

TRAFFIC LIGHT

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1 Problem

The traffic light system contains the following items:

1. The START, STOP buttons; The DAY-NIGHT switch.
2. 3 pairs of W-E lights: GR1, YL1, RD1; 3 pairs of N-S lights: GR1, YL1, RD1.

The process:

1. Select the DAY-NIGHT mode using the switch.
2. Press the START button to start the system.
3. The DAY mode - The system works automatically according to the following cycle length: GR1/RD2-12s — YL1/RD2-3s — RD1/GR2-12s — RD1/YL2-3s.
4. The NIGHT mode - YL1/YL2-3s.
5. Press the STOP button to stop the system immediately.

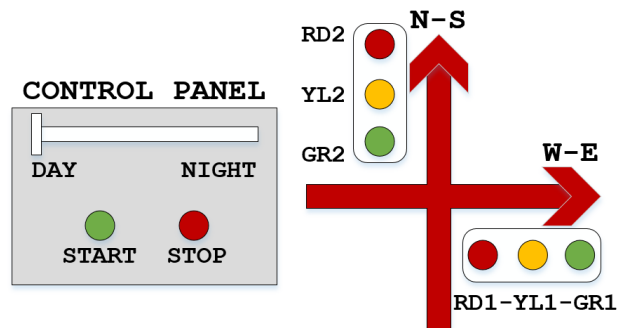


Figure 1: The traffic light system

2 PLC tags

1. INPUT — I0.0:START; I0.1:STOP; I0.2:DAY; I0.3:NIGHT
2. OUTPUT — Q0.0:RD1; Q0.1:YL1; Q0.2:GR1; Q0.3:RD2; Q0.4:YL2; Q0.5:GR2

3 Grafcet

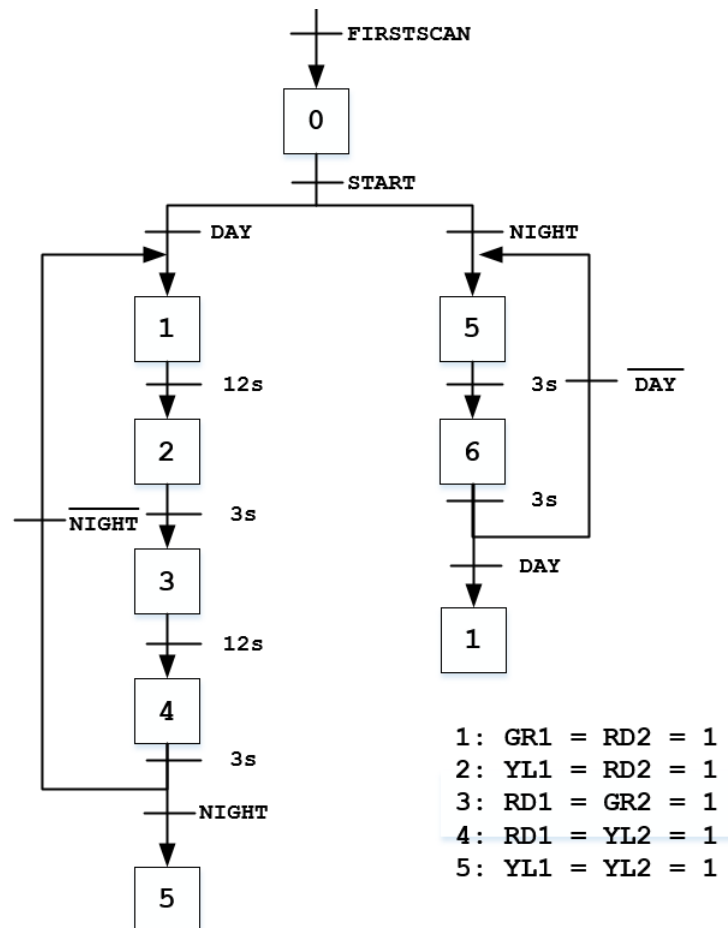


Figure 2: The Grafcet

4 Program

See the attached file (TIA Portal v14 SP1)